Introduction to hypothesis tests

Overview

Last quick review of confidence intervals

A little practice creating confidence intervals in R

Introduction to hypothesis tests

If there is time:

- Hypothesis tests in R
- Hypothesis test terminology

Announcement

Homework 6 has been posted!

The deadline for this homework has been extended to Sunday October 25th at 11pm

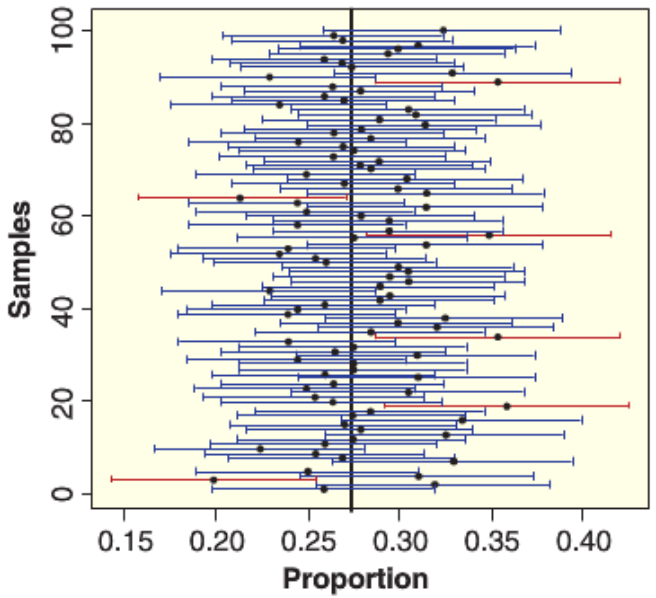
- I.e., the Sunday at the end of fall break
- However, you should be complete it after Thursday's class
  - (so by the original deadline of October 18th)
- Be sure to mark each question on Gradescope!

Very quick review of bootstrap confidence intervals

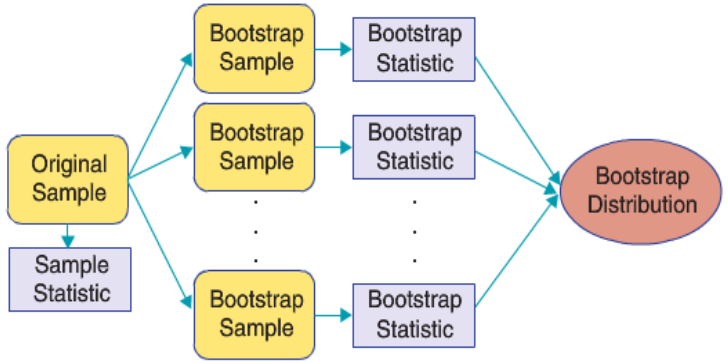
Very quick review: The big picture

Confidence intervals give range of plausible values for a parameter

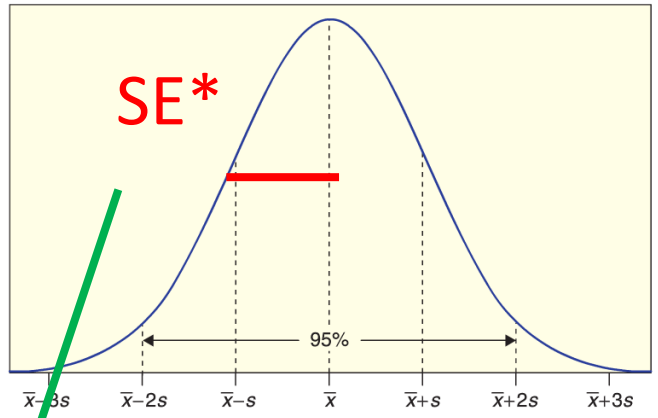
μ







Bootstrap distribution



One sample of data

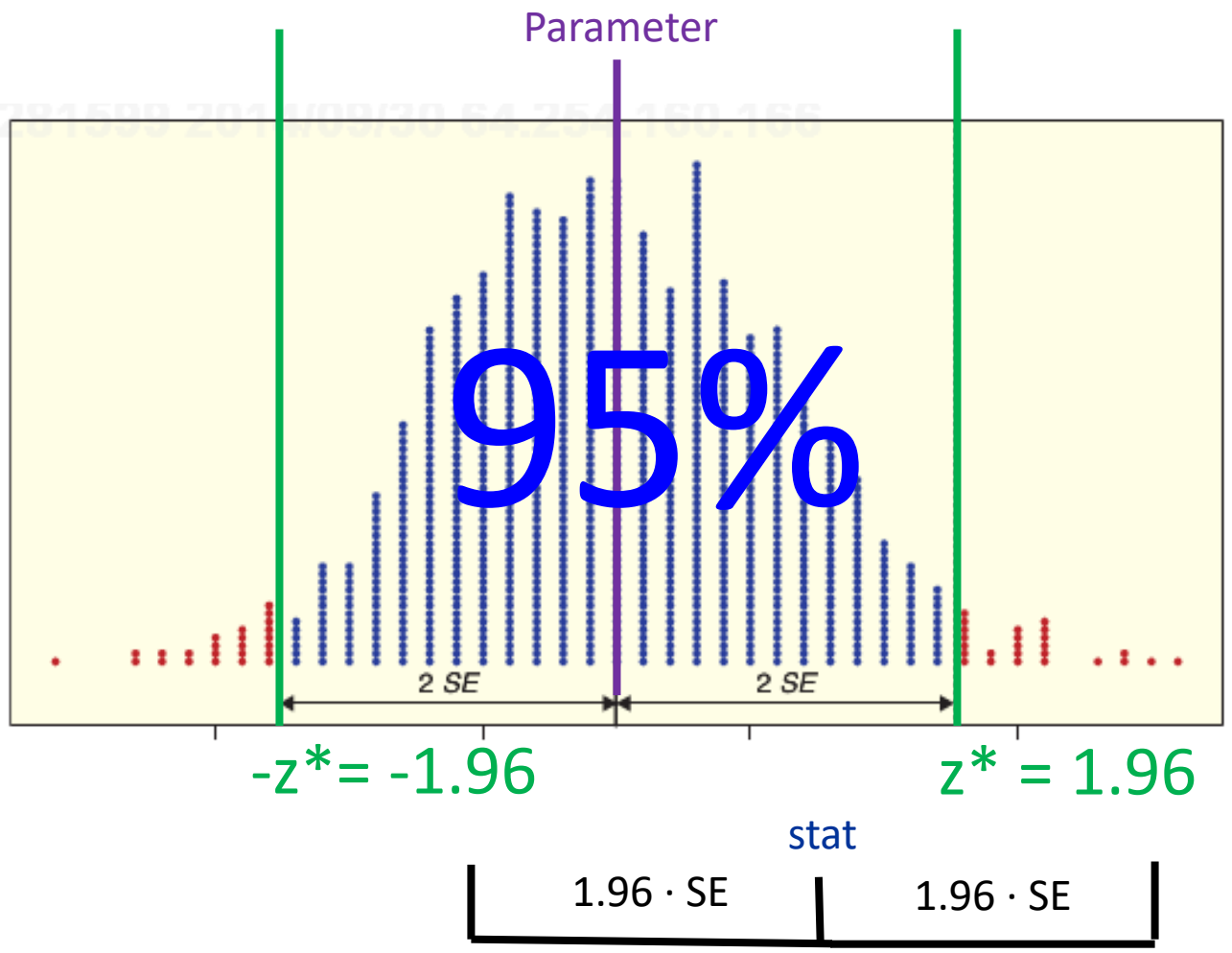
cnorm()

95% confidence interval formula: $\text{stat} \pm z^* \cdot SE^*$

Critical value

For a 95% CI, we choose z* = 2

- 95% of our statistics lie within -2 and 2 SE of the parameter
  - (1.96 to be more precise)




```
z_star <- cnorm(.95)
```

```
1.96
```

Critical value z*

confidence interval: stat ± z* · SE*

CI = stat ± 1.96 · SE*

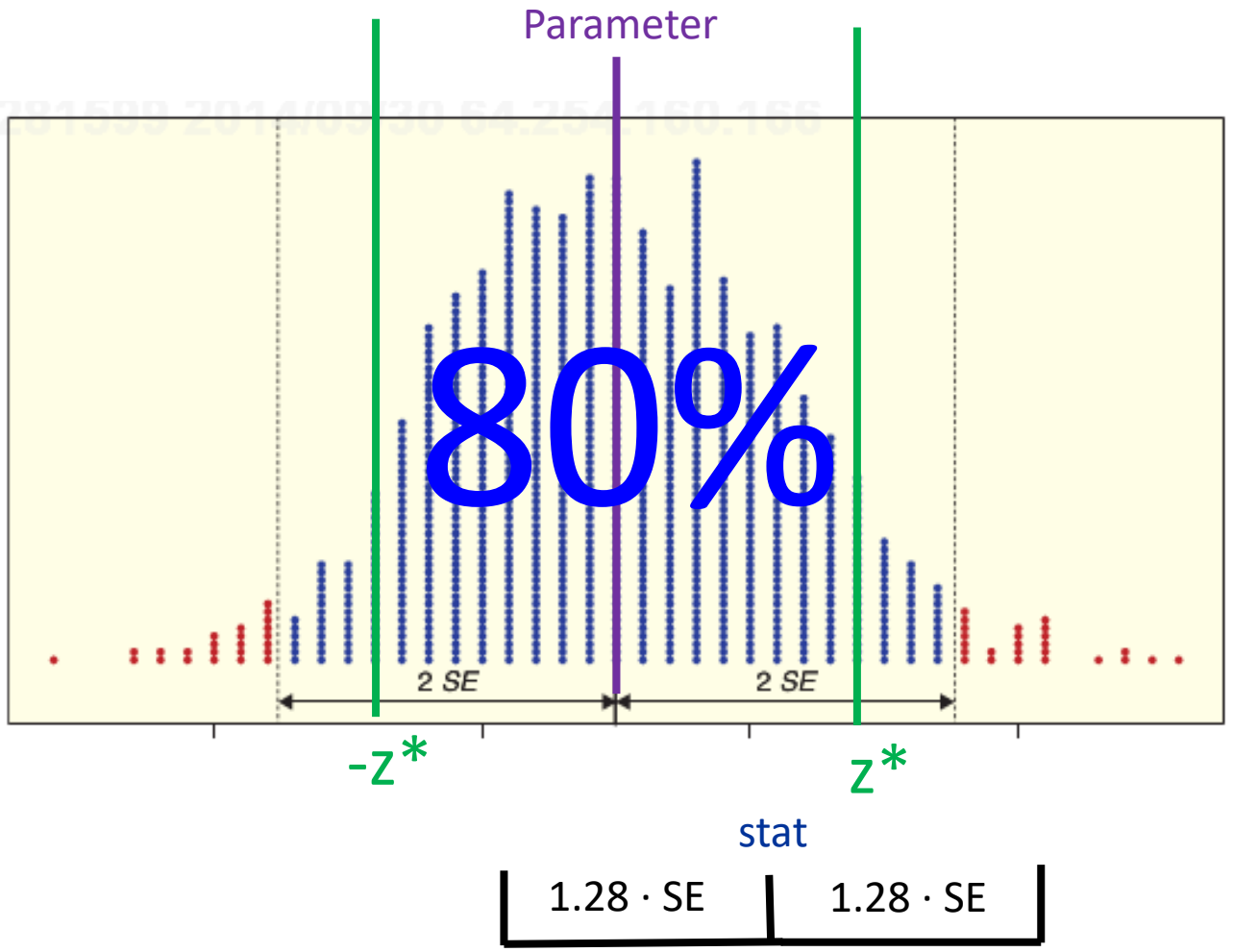
Critical value

We can choose critical values z* for other confidence levels

- E.g. for an 80% CI, we choose z* so that 80% of our statistics lie within -z* and z* SE of the parameter

```
z_star <- cnorm(.80)
```

1.28



Critical value z*

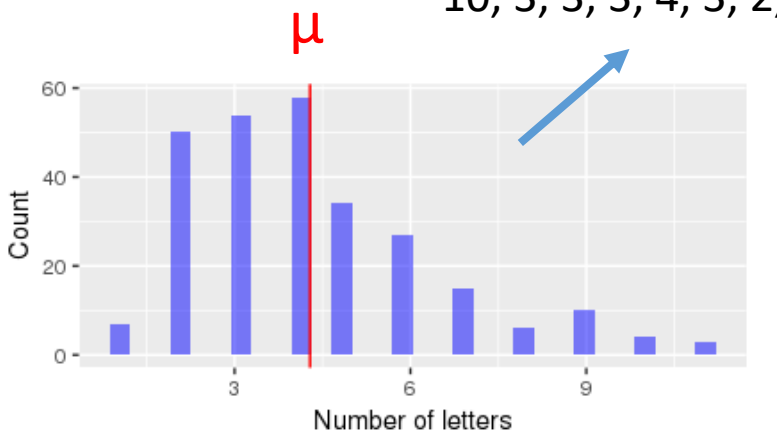
confidence interval: stat ± z* · SE*

CI = stat ± 1.28 · SE*

Review: Using the Bootstrap to get SE*

The sample (n = 10)

10, 3, 3, 3, 4, 3, 2, 6, 4, 5



3, 3, 3, 5, 3,
4, 5, 2, 2, 10

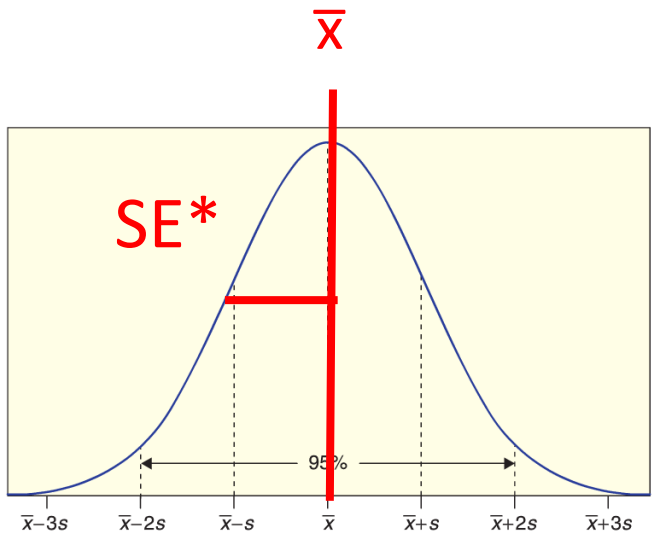
$\bar{x}^* = 4$

3, 3, 2, 3, 6,
4, 6, 5, 3, 6

$\bar{x}^* = 4.1$

5, 3, 2, 3, 3,
3, 10, 3, 4, 3

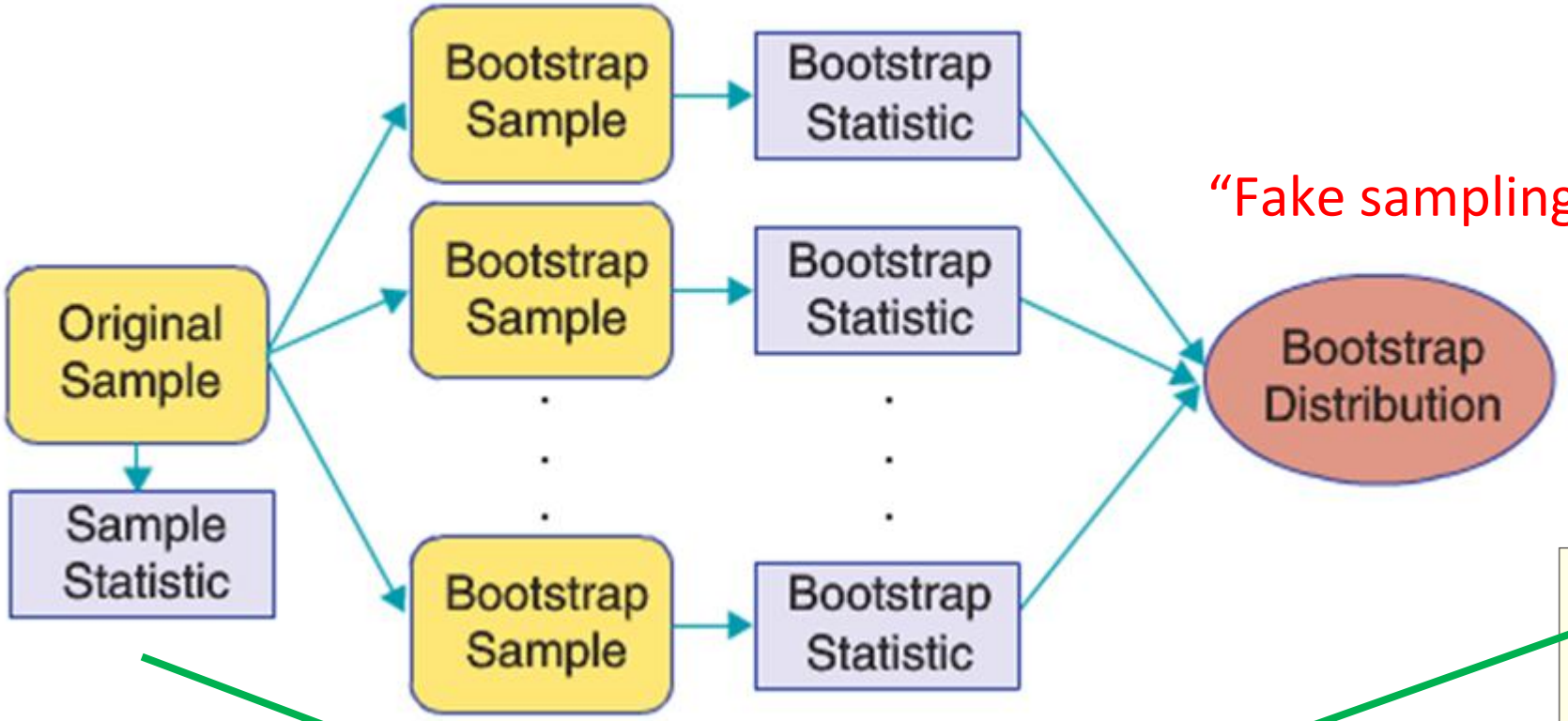
$\bar{x}^* = 3.9$



Bootstrap distribution!

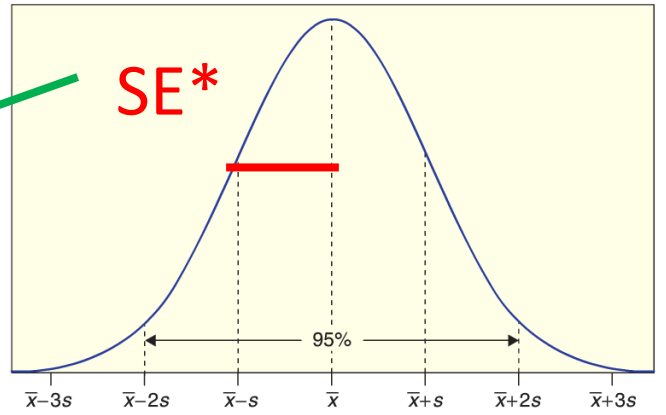
Confidence interval: stat ± z* · SE*

Using the bootstrap to estimate the standard error (SE*)



"Fake sampling distribution"

Bootstrap distribution



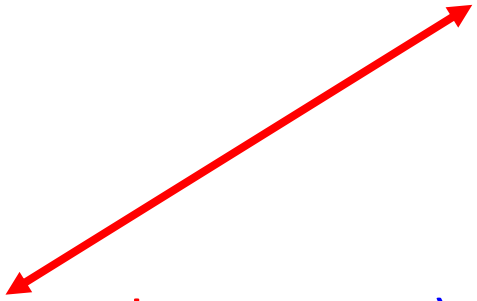
Confidence interval: stat ± z* · SE*

Bootstrap distribution in R for μ

```
my_sample <- c(21, 29, 25, 19, 24, 22, 25, 26, 25, 29)      # n = 10 points here
my_stat <- mean(my_sample)     #  x-bar

bootstrap_dist <-  do_it(10000) * {
        curr_boot <- sample(my_sample , 10, replace = TRUE)
        mean(curr_boot)
}

SE_boot   <-  sd(bootstrap_dist)
z_star <- cnorm(.95)
CI     <-    c(my_stat   -   z_star  * SE_boot,       my_stat   +   z_star  * SE_boot)
```



Confidence interval: $\text{stat} \pm z^* \cdot SE^*$

Bootstrap distribution in R for π

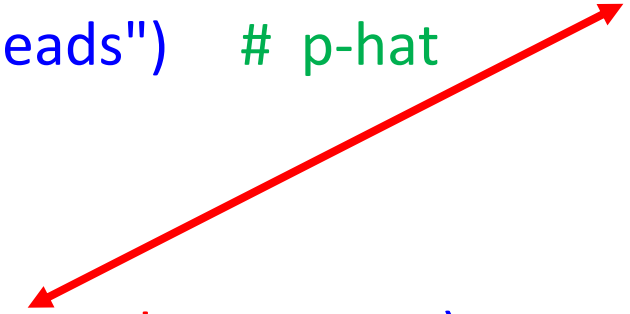
```
my_sample <- c("heads", "tails", "tails", "heads", "heads")     # n = 5
my_stat <- get_proportion(my_sample, "heads")    #  p-hat

bootstrap_dist <-  do_it(10000) * {

        curr_boot <- sample(my_sample , 5, replace = TRUE)
         get_proportion(my_sample, "heads")

}

SE_boot   <-  sd(bootstrap_dist)
z_star <- cnorm(.95)
CI     <-    c(my_stat    -  z_star  * SE_boot,       my_stat    +  z_star  * SE_boot)
```



Confidence interval: $\text{stat} \pm q^* \cdot SE^*$

Shower time

Pereira-Doel et al (2024) conducted a study at the University of Surrey in the United Kingdom examining variables related to water consumption by people taking showers

Let's use the data from this study create a 90% confidence interval for the mean length of time that showers last

Introduction to hypothesis tests

A quick note on probability

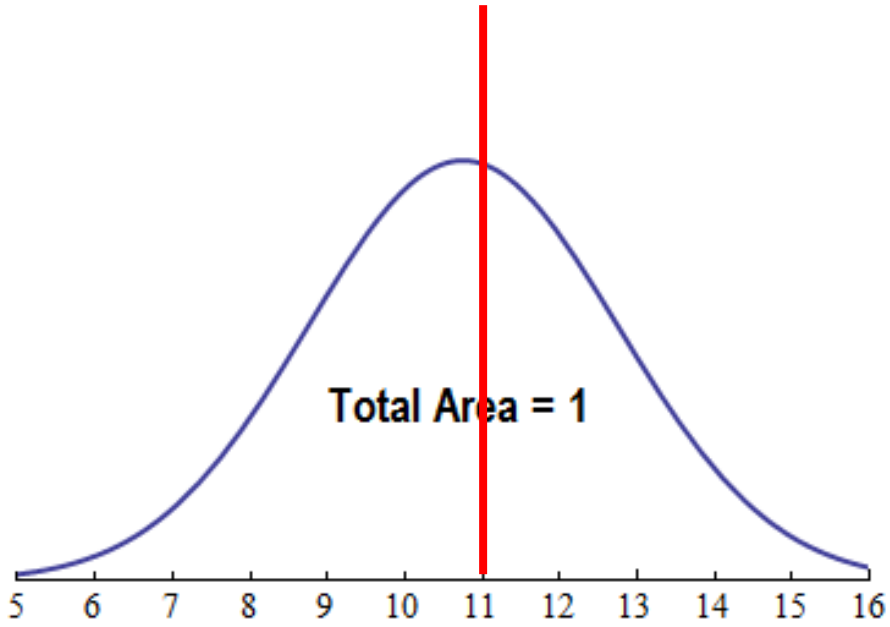
Probability is a way of measuring the likelihood that an event will occur

Probability models assigns a number between 0 and 1 to the outcome of an event (outcome) occurring

We can use a probability model to calculate the probability of an event

For example:

- P(X < 11) = 0.55
- P(X > 20) = 0



Statistical tests (hypothesis test)

A **statistical test** uses data from a sample to assess a claim about a population

Example 1: we might make the claim that Trump's approval rating for all US citizens is 42%

How can we write this using symbols?

- $\pi = .42$

Statistical tests (hypothesis test)

A **statistical test** uses data from a sample to assess a claim about a population.

Example 2: we might make the claim that the average height of a baseball player is 72 inches

How can we write this using symbols?

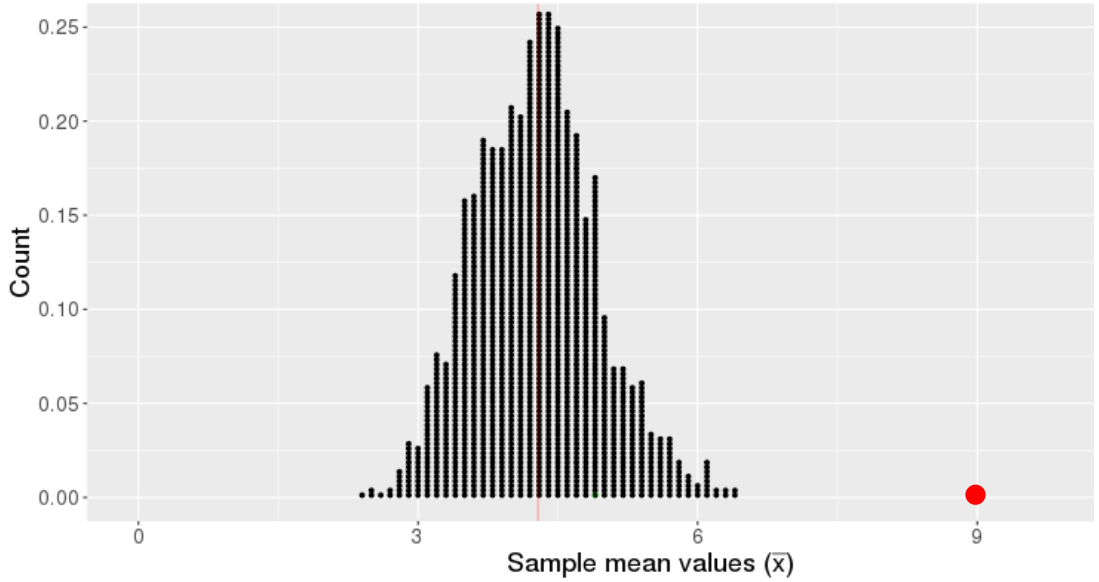
- $\mu = 72$

Basic hypothesis test logic

We start with a claim about a population parameter

- E.g., μ = 4

This claim implies we should get a certain distribution of statistics



If our observed statistic is highly unlikely, we reject the claim

Are dolphins capable of abstract communication?

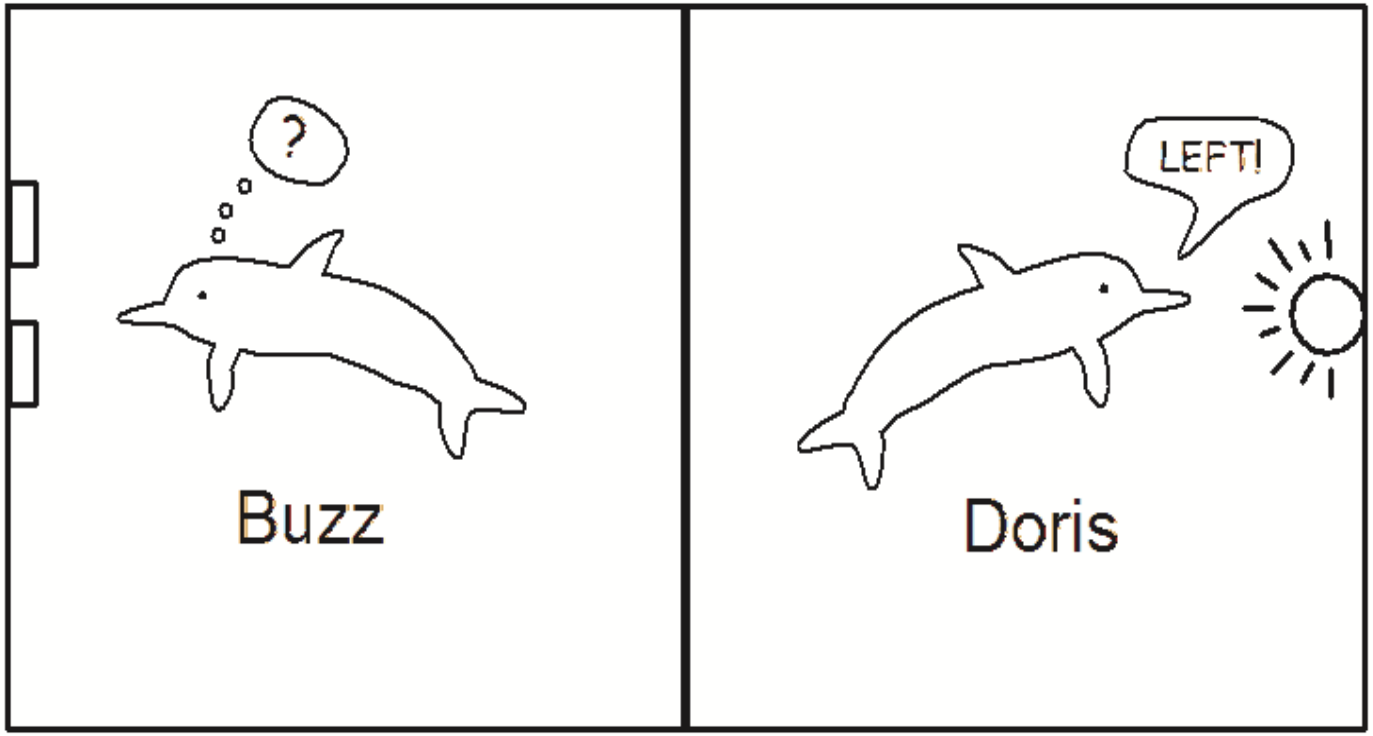
Dr. Jarvis Bastian is the 1960's wanted to know whether dolphins are capable of abstract communication

He used an old headlight to communicate with two dolphins (Doris and Buzz)

- Stead light = push button on right to get food
- Flashing = push button on the left to get food



A canvas was then put in the middle of the pool with Doris on one side and buzz on the other



Left

Right

Buzz got 15 out of 16 trials correct

Please answer the following questions

1. What are the cases here?

2. What is the variable of interest and is it categorical or quantitative?

3. What is the observed statistic - and what symbols should we use to denote it?

4. What is the population parameter we are trying to estimate - and what symbol should we use to denote it?

5. Do you think the results are due to chance?
   - i.e., how many correct answers do you think Buzz would have gotten if he was guessing?

6. Are dolphins capable of abstract communication?

The dolphin communication study

Q: If Buzz was just guessing, what would you expect the value of the parameter to be?

Q: If Buzz was not guessing, what would you expect the value of the parameter to be?

Chance models

How can we assess whether 15 out of 16 correct trials ($\hat{p}$ = .975) is beyond what we would expect to see by chance?

- i.e., beyond what we would expect to see if $\pi = 0.5$?

If buzz was guessing we can model his guesses as a coin flip:

Heads = correct guess

Tails = incorrect guess

Let's flip a coin 16 times and see how many times we get 15 heads

Chance models

To really be sure, how many repetitions of flipping a coin 16 times should we do?

Any ideas how to do this?

Flipping coins using SDS1000 functions

rflip() returns the number of heads out of num_flip coin flips:

```
rflip(num_flips, prob = .5)
```

num_flips: the number of times to flip a coin

- 16 for Doris/Buzz

prob: the probability of success on each trial

- 0.5 if Buzz was guessing

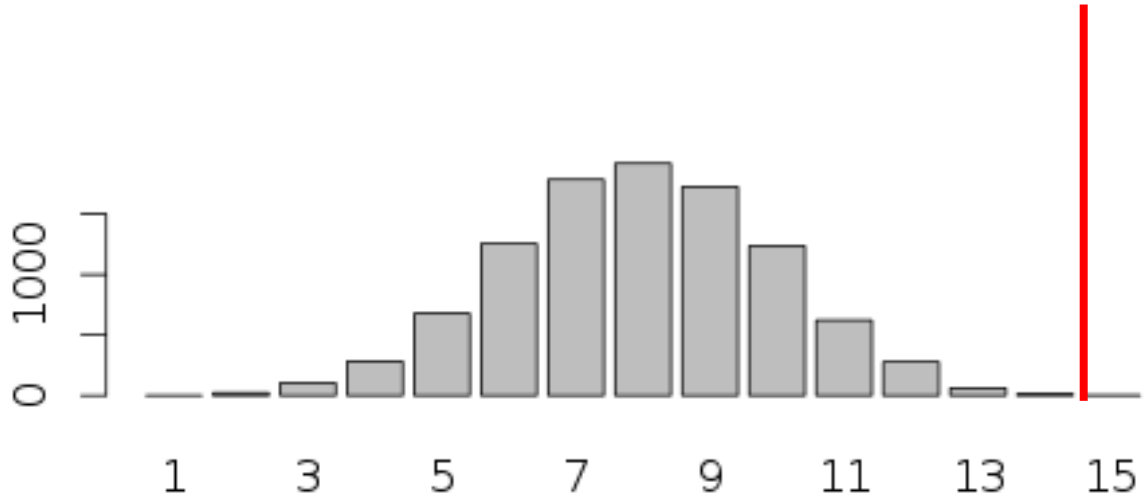
We can repeat flips many times using the do_it() function:

```
library(SDS1000)

flip_simulations  <-  do_it(10000) * {
      rflip(16,  prob = .5)
}
```

Simulating Flipping 16 coins 10,000

0	0
1	1
2	22
3	105
4	283
5	679
6	1257
7	1786
8	1920
9	1726
10	1238
11	623
12	279
13	63
14	15
15	3
16	0

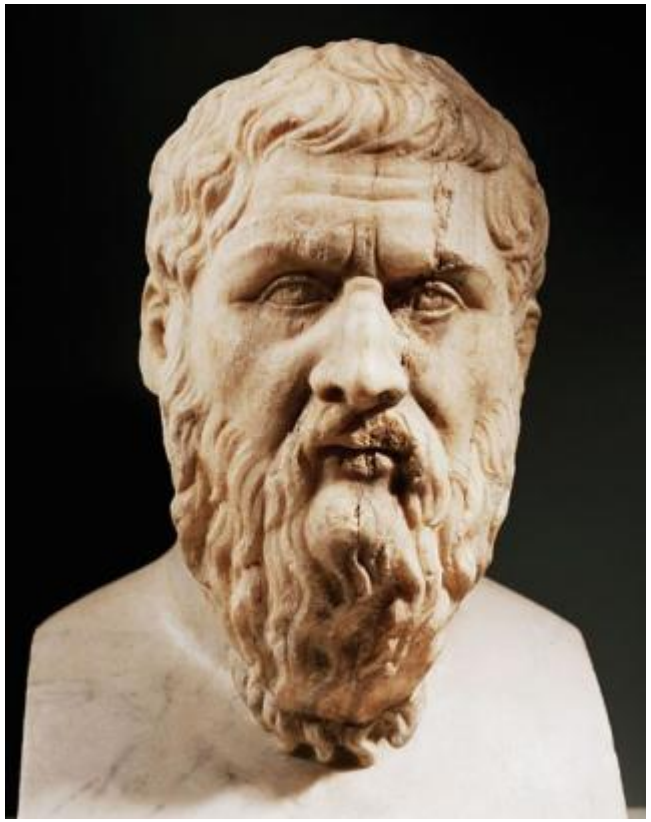


Q: Is it likely that Buzz was guessing?

Q: Are dolphins capable of abstract communication?

Hypothesis tests: central ideas and terminology

Question: who is this?



G believes in the ***null hypothesis***

- that Buzz was guessing

How can we write the null hypothesis in symbols?

$H_0: \pi = 0.5$

We believe in the ***alternative hypothesis***

- Doris an Buzz can communicate

How can we write the alternative hypothesis in symbols?

$H_A: \pi > 0.5$

How can we convince Gorgias?

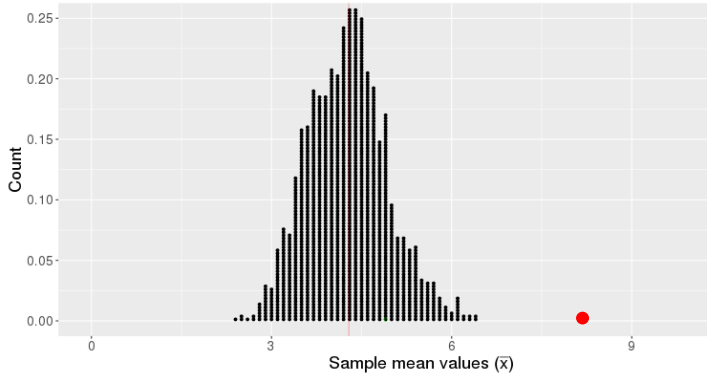
To prove Gorgias wrong, we will start by assuming he is right!

Namely, we will assume H_0 (that $\pi = 0.5$)

We will then generate a number of statistics ($\hat{p}$) that are consistent with H_0

- i.e., we will create a ***null distribution***

If our observed statistic looks very different from the statistics generated under we can reject H_0 and accept H_A

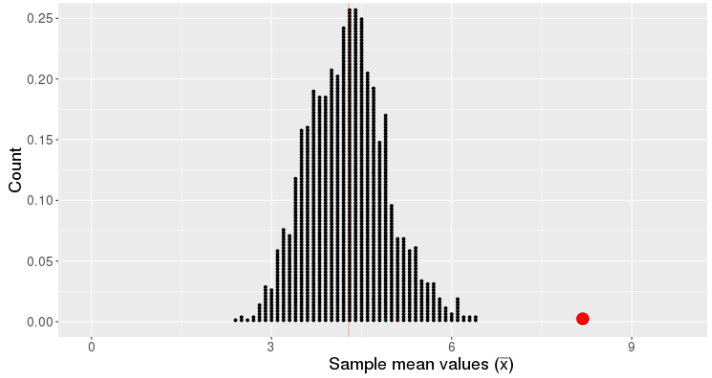


Terminology

Null Hypothesis (H_0): Claim that there is no effect or no difference

Alternative Hypothesis (H_a): Claim for which we seek significant evidence

The alternative hypothesis is established by observing evidence that inconsistent with the null hypothesis



Review: the dolphin communication study

1. What is the null hypothesis in words?

2. We can write this in terms of the population parameter as:

$H_0: \pi = 0.5$

3. What is the alternative hypothesis?

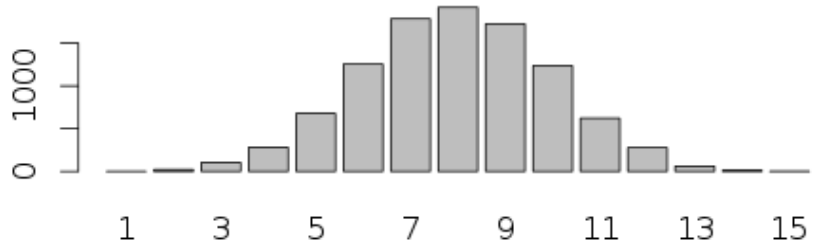
$H_A: \pi > 0.5$

Null Distribution

A **null distribution** is the distribution of statistics one would expect if the null hypothesis (H_0) was true

i.e., the null distribution is the statistics one would expect to get if nothing interesting was happening

- Note: the Lock5 textbook calls this the "randomization distribution"



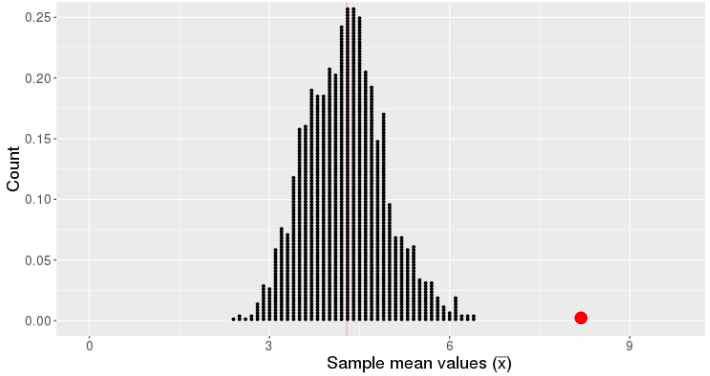
P-values

A **p-value** is the probability, of obtaining a statistic as as (or more) extreme than the observed sample *if the null hypothesis was true*

- i.e., the probability that we would get a statistic as extreme as our observed statistic from the null distribution

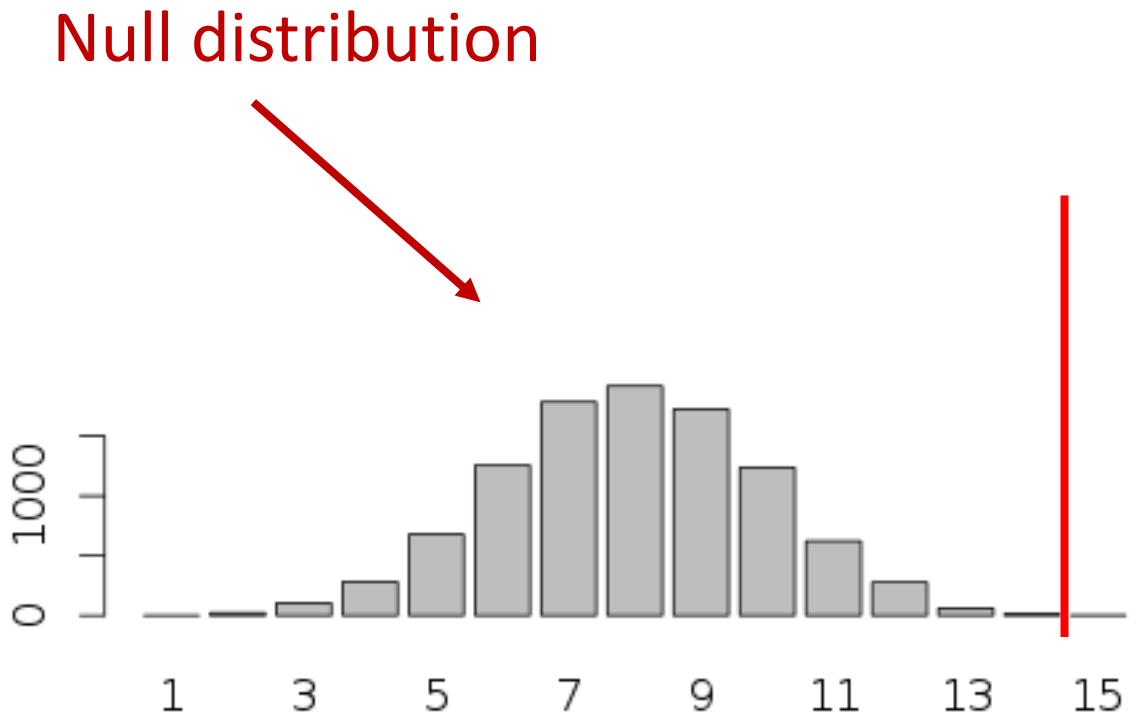
$$P(\text{STAT} \geq \text{observed statistic} \mid H_0 = \text{True})$$

The smaller the p-value, the stronger the statistic evidence is against the null hypothesis



Buzz and Doris example

0	0
1	1
2	22
3	105
4	283
5	679
6	1257
7	1786
8	1920
9	1726
10	1238
11	623
12	279
13	63
14	15
15	3
16	0



p-value = 3/10000 = 0.0003

Statistical significance

When our observed sample statistic is unlikely to come from the null distribution, the results are called "**statistically significant"**

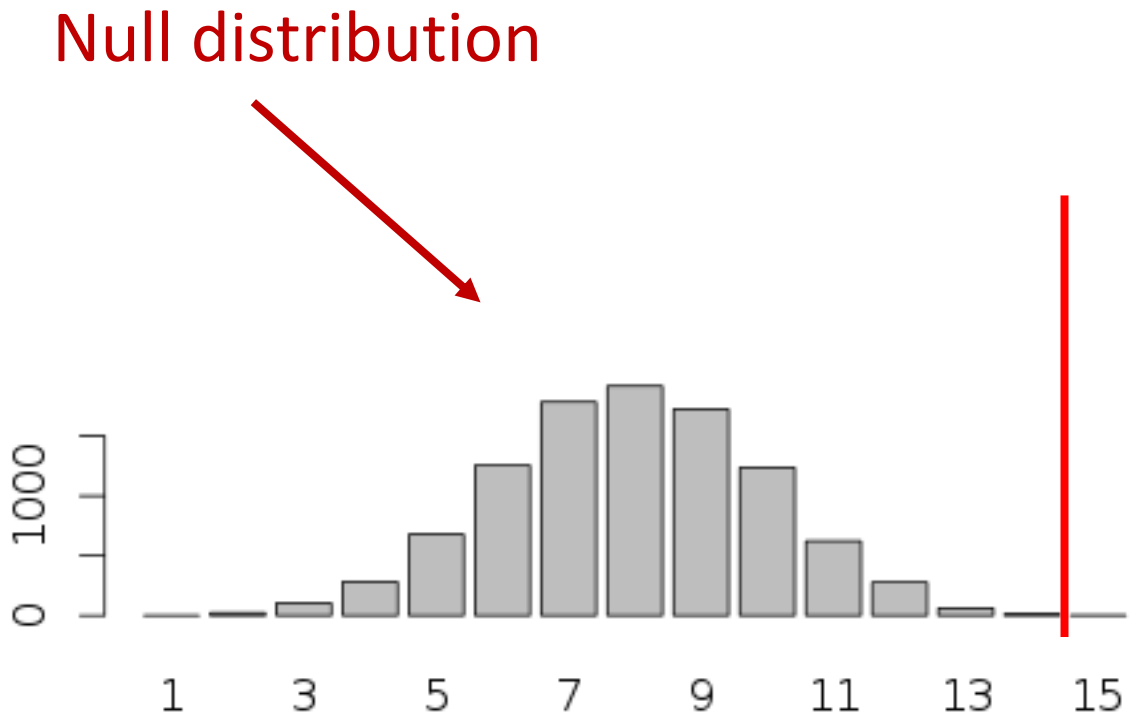
- i.e., we have a small p-value

'Statistically significant' results mean we have strong evidence against H_0 in favor of H_A

Side note: We will discuss whether we should really use the term "statistical significance" later in the semester

Buzz and Doris example

0	0
1	1
2	22
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16	0

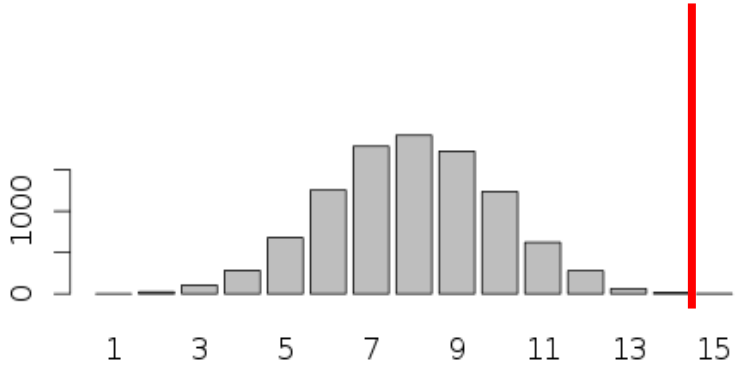


p-value = 3/10000 = 0.0003

Getting p-values using SDS1000 functions

Flipping coins many times:

```
flip_simulations  <-  do_it(10000) * {
      rflip(16,  prob = .5)
}
```



We can get the proportion of values as or more extreme than an observed statistic (obs_stat) using the ptail() function:

```
obs_stat <-  15
p_value  <- ptail(obs_stat,  flip_simulations,  lower.tail = FALSE)
```

Key steps hypothesis testing

1. State the null hypothesis… and the alternative hypothesis
   - Buzz is just guessing so the results are due to chance: $H_0: \pi = 0.5$
   - Buzz is getting more correct results than expected by chance: $H_A: \pi > 0.5$

2. Calculate the observed statistic
   - Buzz got 15 out of 16 guesses correct, or $\hat{p} = .973$

3. Create a null distribution that is consistent with the null hypothesis
   - i.e., what statistics would we expect if Buzz was just guessing

4. Examine how likely the observed statistic is to come from the null distribution
   - What is the probability that the dolphins would guess 15 or more correct?
   - i.e., what is the p-value

5. Make a judgement
   - If we have a small p-value, this means that $\pi = .5$ is unlikely and so $\pi > .5$
   - i.e., we say our results are 'statistically significant'